Community Composting 101

Site Design & Management

Neighborhood Soil Rebuilders
COMPOSTER TRAINING PROGRAM

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INSTITUTE FOR Local Self-Reliance

Site Design & Management

- ✔ Navigating State & Local Permitting & Regulations
- ✔ Choosing a System
- ✔ Site Layout & Management Plan
- ✔ Health & Safety
- ✔ Tips to Avoid Rodents
- ✔ What Do You Notice?

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Permitting & Regulations

✔ **Zoning – all zoning is local**


Ohio Environmental Protection Agency

Urban Agriculture, Composting and Zoning

A zoning code model for promoting composting and organic waste diversion through sustainable urban agriculture

https://www.epa.ohio.gov/portals/34/document/guidance/GD%201011_UrbanAgCompostingZoning.pdf

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Permitting & Regulations

✔ **Zoning – all zoning is local**

✔ **State permitting rules**

- Exemptions typical for on-farm and small-scale sites


On-Farm Composting Rules and Permit Exemptions

https://ilsr.org/rule/on-farm-composting/

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Permitting & Regulations

- **Zoning – all zoning is local**
- **State permitting rules**
  - Exemptions typical for on-farm and small-scale sites

> TN Dept. of Environment & Conservation: Compost Site Permit Exemptions
>
> Site is allowed to process without a permit:
>
> 50 cubic yards/year of putrescible material without a permit (with no restrictions on use) using traditional means like windrows and piles
>
> 100 cubic yards/year in an in-vessel system

Permitting & Regulations

- **Zoning – all zoning is local**
- **State permitting rules**
  - Exemptions typical for small-scale sites
  - May not be exempt if distributing compost

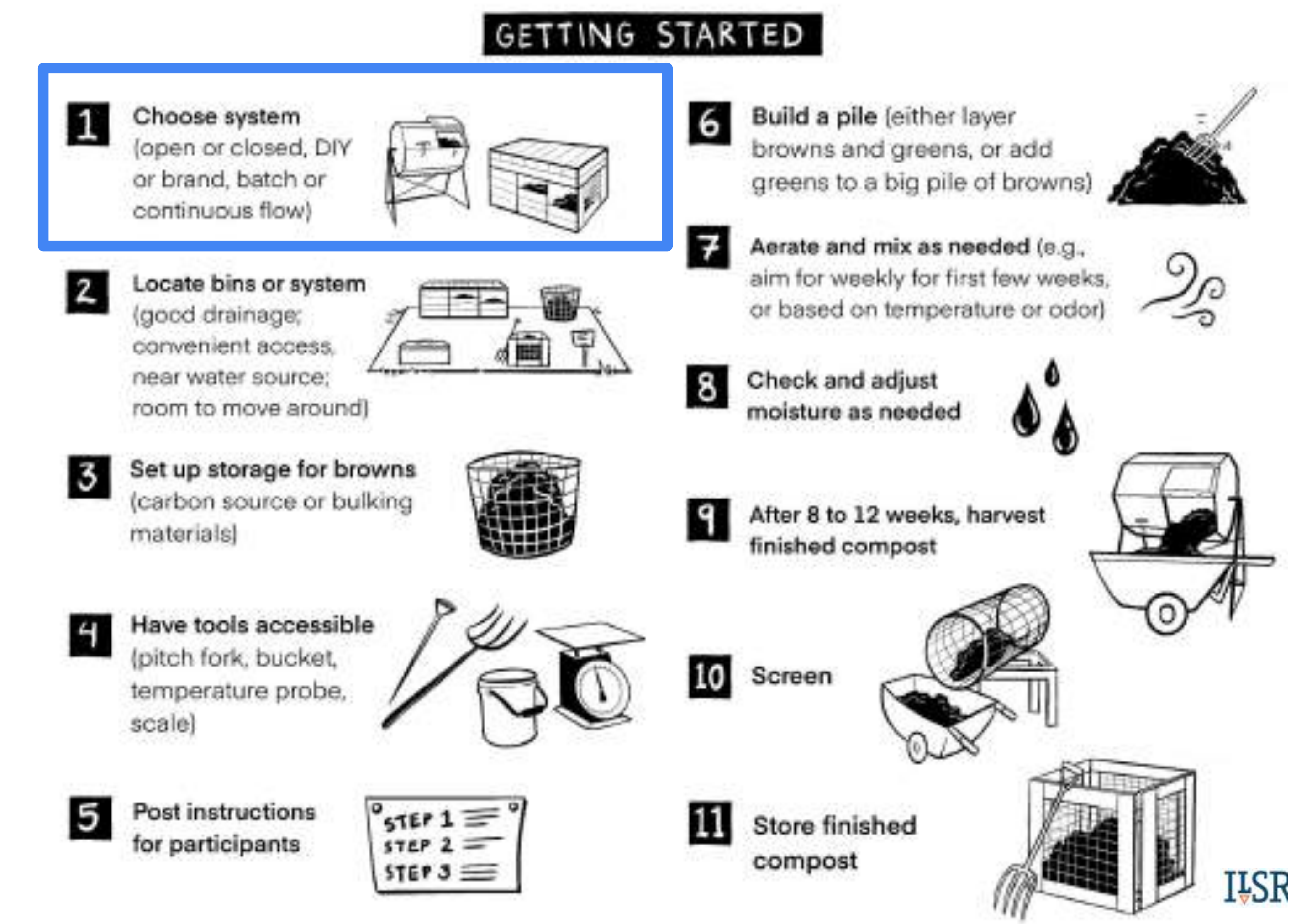
https://mda.maryland.gov/plants-pests/Documents/New-Soil-Conditioner-Registration-Form.pdf

Permitting & Regulations

- ✔ **Zoning – all zoning is local**
- ✔ **State permitting rules**
  - Exemptions typical for small-scale sites
  - May not be exempt if distributing compost
- ✔ **One resource: CompostLaw.Org**
- ✔ **Contact your regulators**

Planning your compost site

https://ilsr.org/composting-bmp-guide/


GETTING STARTED
1 Choose system (open or closed, DIY or brand, batch or continuous flow)
2 Locate bins or system (good drainage; convenient access, near water source; room to move around)
3 Set up storage for browns (carbon source or bulking materials)
4 Have tools accessible (pitch fork, bucket, temperature probe, scale)
5 Post instructions for participants
STEP 1
STEP 2
STEP 3
6 Build a pile (either layer browns and greens, or add greens to a big pile of browns)
7 Aerate and mix as needed (e.g., aim for weekly for first few weeks, or based on temperature or odor)
8 Check and adjust moisture as needed
9 After 8 to 12 weeks, harvest finished compost
10 Screen
11 Store finished compost
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Many compost systems!
NEW DEAL CAFE
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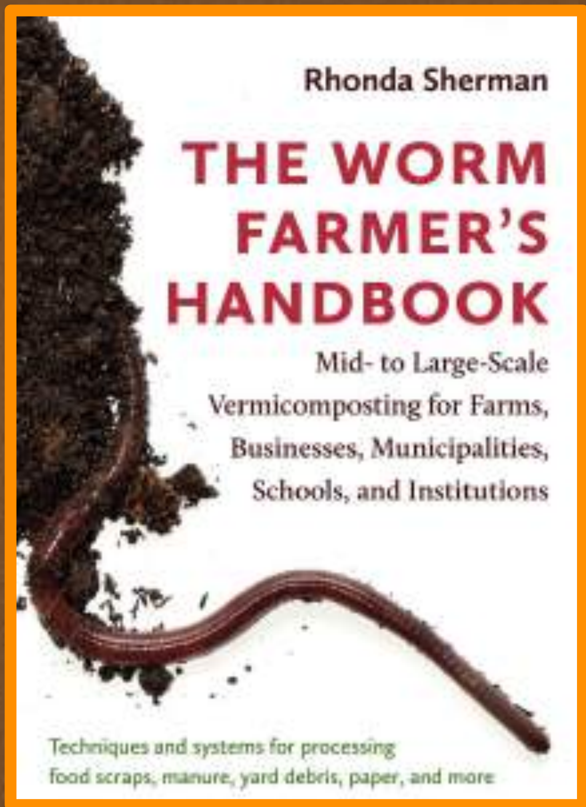

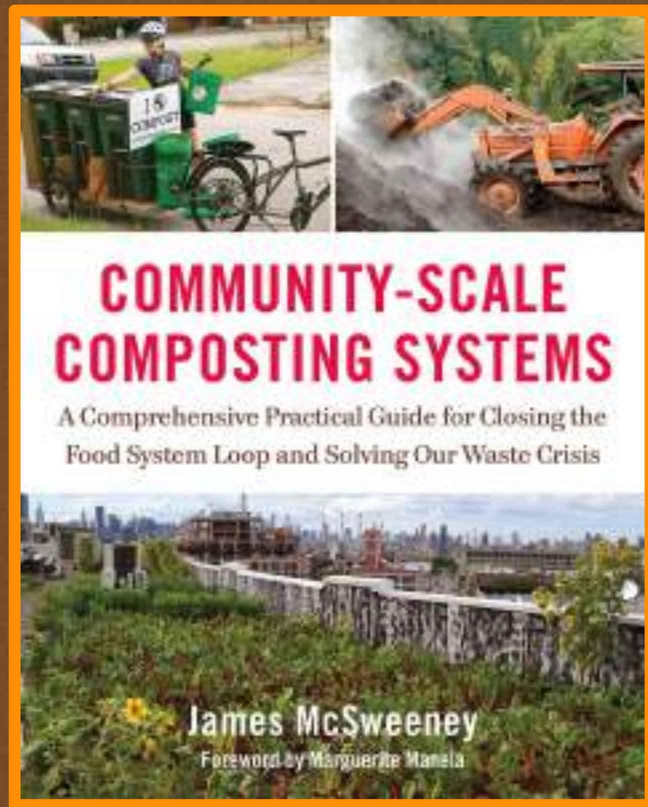

https://ilsr.org/webinar-vermicomposting-january-2019/

https://ilsr.org/webinar-community-scale-systems-march-2019/

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Webinar: Community-Scale Composting Systems with James McSweeney

Clear understanding of capacity is key to <u>business planning</u> and <u>viability</u>

https://ilsr.org/webinar-community-scale-systems-march-2019/

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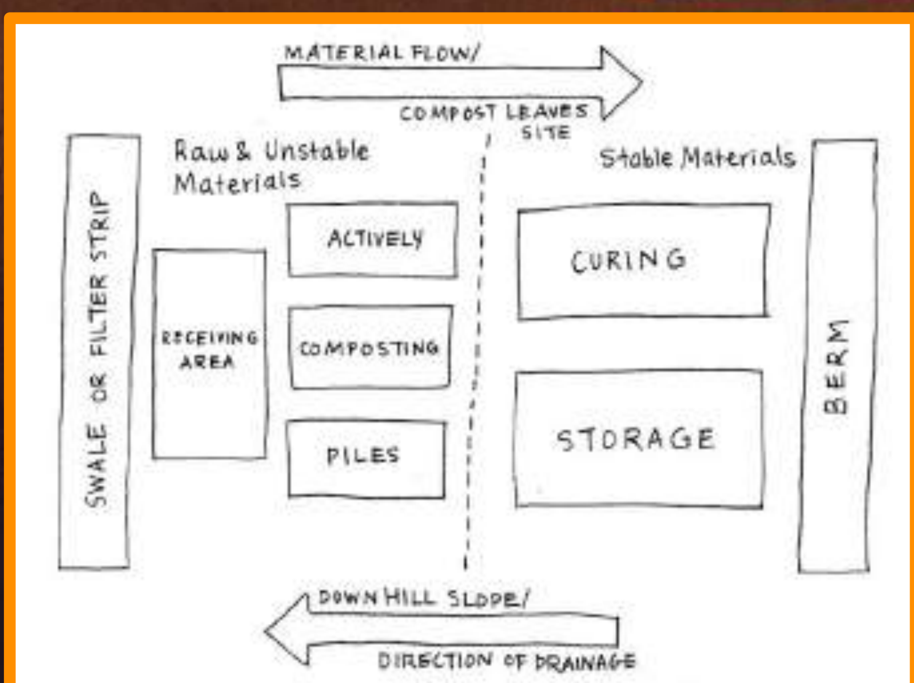
Some other key questions

- What materials will you process and from where will you source them?
- What is the estimated volume and weight of materials?
- Who will be in charge of managing the composting process?
- What kind of budget do you have for equipment?
- How much space do you have?
- How much space do you need?
- How close and potentially sensitive are your neighbors?
- If you plan to use the compost on-site, how much do you need?
- If you want to sell your compost, what volume is your local market likely to buy?
- Are you going to provide compost to organic growers?

Source: L. Bilsens Brolis, B. Platt, Community Composting Done Right: A Guide to Best Management Practices, Institute for Local Self-Reliance, 2019. (www.ilsr.org/composting-bmp-guide).

Site Layout: Set yourself up for success!



The layout of a composting site should allow for a logical and efficient flow of materials and people, while minimizing the potential for contamination of more stable materials. Adapted from Oregon State University, Agricultural Composting and Water Quality (2013).

Drainage Is Important

- ✔ Put finished product uphill of non-finished product
- ✔ Do not let "contact" water runoff or drain into streams or other surface water
- ✔ Don't want standing water onsite

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Prospect Heights Community Farm (NYC) Video

https://youtu.be/rS2m9j_7hAs

Note: watch this video and notice:

1. Site set-up and flow
2. Signage
3. Data recording
4. Browns storage
5. Compost curing
6. Finished compost storage

Photo: ILSR

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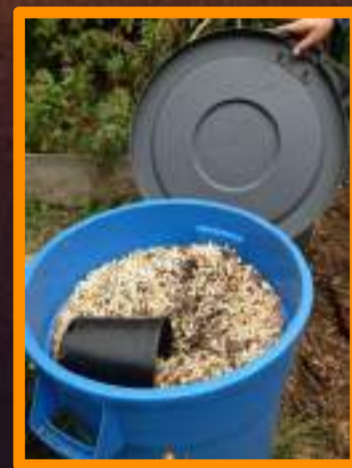
Secure Adequate Carbon

Carbon storage at Battery Park, NYC

DC community cooperative site



Wood shavings at Prospect Heights Community Garden, Brooklyn

Photos ILSR

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Examples of Carbon Storage

Carbon storage at BK ROT, NYC

Howard Univ.

Austin community site

Red Hook Community Farm Compost Operation, Brooklyn, NYC

Photos ILSR

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How will material be collected or received?

Real Food Farm, Baltimore

Red Hook Community Farm, NYC

Community Compost Co., NY

Photos: ILSR (top left and right), Community Compost Co. (right bottom)

- ✔ No physical site or drop offs
- ✔ One drop point that you pop-up or manage, where people bring scraps
- ✔ Multiple drop points
- ✔ Working with a community garden: you accept drop offs
- ✔ Working with a community garden: you make compost
- ✔ Working with a community garden: other people make the compost
- ✔ Cooperate with other sites – such as farms, commercial compost sites – where you drop off organics that you have gathered from others

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Photos: ILSR

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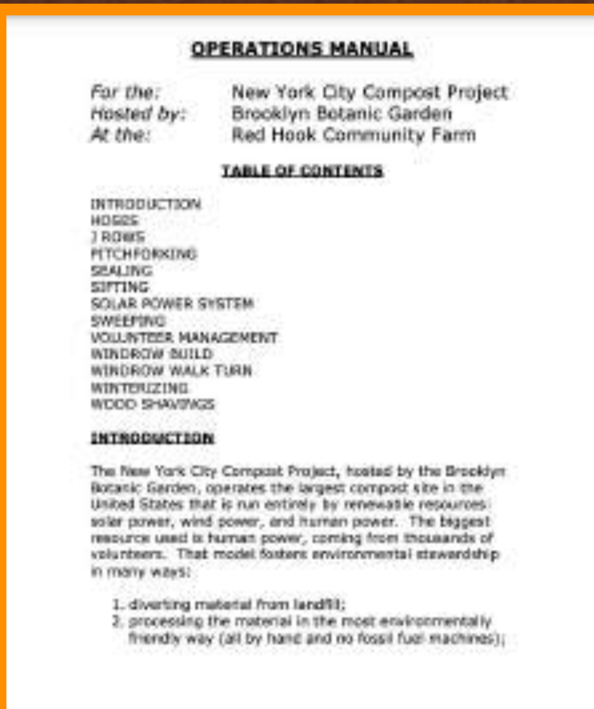
Identify Your Team

Members of the Greenbelt Maryland Zero Waste Circle, an all-volunteer group. *Photo: ILSR*

Identify Your Management Plan



OPERATIONS MANUAL

For the: New York City Compost Project
Hosted by: Brooklyn Botanic Garden
At the: Red Hook Community Farm

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INTRODUCTION

The New York City Compost Project, hosted by the Brooklyn Botanic Garden, operates the largest compost site in the United States that is run entirely by renewable resources: solar power, wind power, and human power. The biggest resource used is human power, coming from thousands of volunteers. That model fosters environmental stewardship in many ways:

1. diverting material from landfill;
2. processing the material in the most environmentally friendly way (all by hand and no fossil fuel machines);

Red Hook Community Farm Compost Operation, Brooklyn, NYC

Photos: ILSR

Red Hook Community Farm

Photos: ILSR

Concrete Pad

Source: @RedHookCompost

Site surface / compost pad is important

Baltimore Compost Collective

Photos: ILSR

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East Capitol Urban Farm, District of Columbia

The site is laid out in a linear pattern: feedstocks enter on one side, and finished compost exits the other.

Photos: ILSR

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Gravel pad example

Photos: ILSR

Hidden Harvest Community Farm, Baltimore   Photos: ILSR

How will you mix?

Photos: ILSR

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Earth Matter's Mixer & Grinder

Earth Matter, Governors Island, NYC

Photos: ILSR

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How will you sift or screen?

Sifters at Earth Matter's compost demonstration site, Governors Island, NYC. Photos: Earth Matter

Sifter at Earth Matter's compost demonstration site, Governors Island, NYC. Photo: ILSR

Trommel Screener

Trommel screener building at Malcolm House in Baltimore. *Photo: ILSR*

Custom built trommel screener at ECO City Farms in Bladensburg, MD. *Photo: ILSR*

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W.B. Saul High School, Philadelphia

Farm scale screener.

Photos: ILSR

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Earth Matter,
Governors Island, NYC

Sittler screener *Photo: ILSR*

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Storing Finished Compost -
Howard Univ. Garden site

Photos: ILSR

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Storing Finished Compost –
W.B. Saul High School

Photo: ILSR

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Storing Finished Compost –
Red Hook Community Farm

Photo: ILSR

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Health & Safety

Follow basic health and safety hygiene:

- Use gloves appropriately
- Careful with hand-to-mouth contact
- Wash hands
- Treat cuts and scrapes immediately
- Protect wounds

Airborne Particles

- Airborne particles can cause physical irritation (inhalable)
- Main concern: *Aspergillus fumigatus* (fungal spores)
- Consider wearing a mask for dry, dusty conditions and if you have asthma, other respiratory issues, or cystic fibrosis (N95 mask recommended; any face covering is better than none)

Source: U.S. Composting Council, Foundations of Composting Workshop

A special word about rodents

Site the compost system away from any other "harborage."

Rodents can carry diseases including hantavirus and leptospirosis.

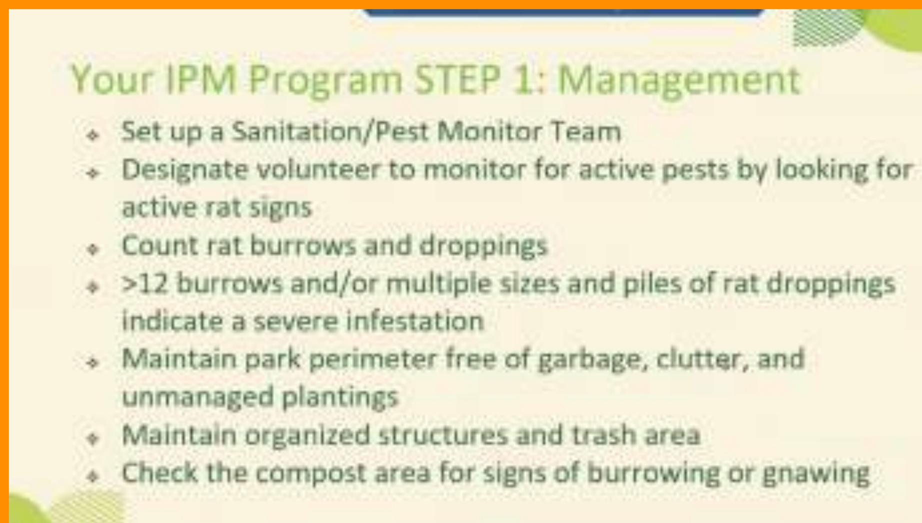
Leptospirosis is a rare bacterial infection. It's spread through their urine.

One healthy female Norway rat can have 84 offspring in 1 year.

You must beat back rodent populations.

NYC Rat Academy

https://www1.nyc.gov/site/doh/services/rats-control-training.page



Your IPM Program STEP 1: Management

- Set up a Sanitation/Pest Monitor Team
- Designate volunteer to monitor for active pests by looking for active rat signs
- Count rat burrows and droppings
- >12 burrows and/or multiple sizes and piles of rat droppings indicate a severe infestation
- Maintain park perimeter free of garbage, clutter, and unmanaged plantings
- Maintain organized structures and trash area
- Check the compost area for signs of burrowing or gnawing

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9 TIPS TO AVOID RODENTS & OTHER UNWANTED CRITTERS

1 Do not compost meat, dairy, fats/oil, cooked food.

2 Incorporate all bits of food well into pile, never leave any exposed or visible.

3 Cover all piles with a thick layer of leaves or other browns.

4 Maintain at least 3 feet of open space all around your system (this makes rodents nervous about predators).

5 Avoid clutter and trim back grasses and shrubs to eliminate potential rodent hiding places.

6 Turn piles thoroughly and regularly so rodents see no opportunity for a habitat.

7 Bin systems need a barrier at the base to prevent habitat formation where it's nice and warm (1/4-inch hardware cloth or something else inhospitable like cement or a 6-inch dug-out pit with sand or gravel).

8 If you have space, consider moving system from time to time. Rodents like habitats that are undisturbed.

9 Activity is good!

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What do you notice?

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What will your plan be?

Getting Started & Engaging the Community

- ✔ **Getting Started**
  - Clarifying goals
  - Collection vs. composting
  - Entity structure
- ✔ **Engaging the Community**
  - Signage
  - Managing volunteers
- ✔ **Parting Thoughts**

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